



Multi-instance multi-label position-aware doubly graph convolutional networks

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Abstract

Multi-instance multi-label learning is a general framework in which each sample is represented as a bag of instances associated with multiple labels. However, two weaknesses remain and hinder its performance in real-world tasks. One is that bag generators often neglect positional information (e.g., the location of a pixel in the image) when generating bags, making position-related labels indistinguishable. The other is that the MIL assumption does not always hold. In some real-world tasks, labels have hierarchical low-level concepts, and these concepts are related to certain combinations of instances instead of one single instance. In this paper, we propose the Position-Aware Doubly Graph Convolutional Networks (**padGCN**). On the one hand, **padGCN** generates bags by arranging instances in a multi-instance graph to aggregate instances' features by exploiting positional relationships among them. Then instances that aggregate other instances' features are input into a neural network to obtain sub-sub-concepts used for multi-label learning. On the other hand, **padGCN** learns sub-concepts from labels and organizes sub-sub-concepts, sub-concepts, and labels in a tripartite multi-label graph in hyperbolic space to exploit their hierarchical structure. Experiments are conducted on 6 image and text data sets. Compared to the SOTA methods, padGCN averagely achieves 5% improvement on 7 measurements. Pair-wise t-test results on 42 experiments indicate that padGCN is significantly better than SOTA methods in 30 experiments, comparable to SOTA methods in 12 experiments, and never worse than SOTA methods, which verifies the superiority and robustness of padGCN. Runtime experiments show that padGCN is comparable to SOTA methods and is computationally efficient.

Keywords

multi-instance; multi-label; GCN

1 Introduction

Multi-Instance Multi-Label (MIML) [1] learning is a generalization of the single-instance single-label learning, in which each sample is represented as a bag of instances associated with multiple labels. Due to this generality, many real-world tasks can be formalized as MIML learning problems such as image classification [2–4] and text classification [5,6].

From the aspect of multi-instance learning, many methods follow the standard assumption [7] that instances are independent and identically distributed (IID) and labels are triggered by key instances. Methods like MIMLSVM [1] convert multiple instances into a single one to transform it into a multi-label learning problem. However, this standard MIL assumption cannot be satisfied in many real-world scenarios, and a new assumption that instances are not independent and the combination of instances triggers the bag label [8].

One feasible method is to use GCN to aggregate instances' features and learn such combination [9]. In recent years, instances' positions have been noticed in some scenarios for exploring their relationships

and combinations. In a *Drosophila* gene expression pattern annotation task, instances are found associated with the position where it is located in the image [10]. Another research calculates the distances using instances' position in Mahalanobis space between bags, which improves the performances of protein function prediction tasks [11]. However, the instances' positions have not been used as an important feature. Besides, the methods above neglect the importance of instances' positions in aggregating instance features to generate instance combinations. Positional features have great influence in aspects like instance combination. Here is an example to show such importance. An image showing that the sun is near the horizon is labeled *morning* or *dawn*, and another one showing that the sun is far from the horizon is labeled *noon*. A similar phenomenon occurs in other scenarios like text classification tasks.

For the multi-label part of MIML learning, methods like MIMLBoost [1] following MIL assumption reduce one multi-label problem into multiple binary classification ones and simply combine